LOGISTIC REGRESSION

A study of customer churn rate using their consumption behavior as predictors in E-commerce

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1. Introduction

1.1. Database

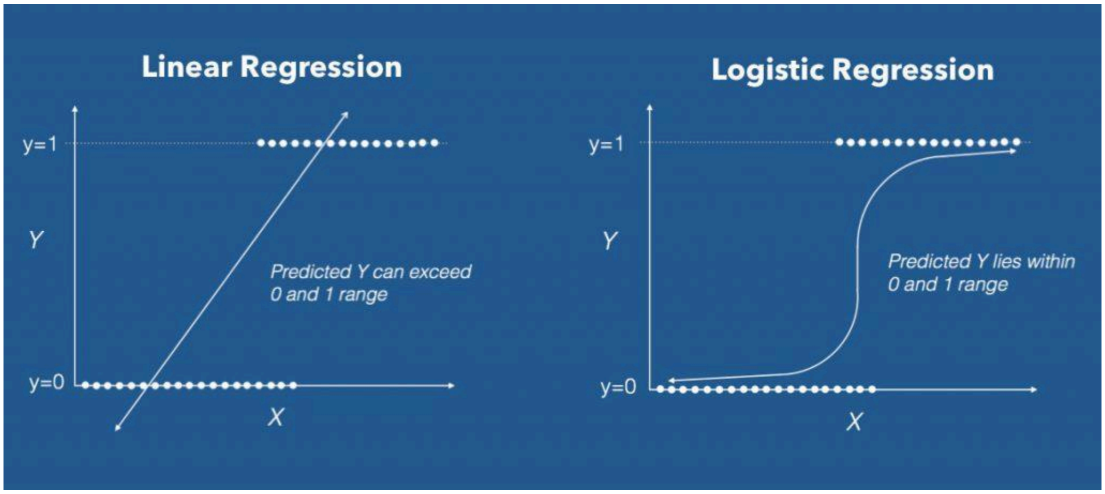
Dataset: Multi-Channel E-commerce with Behavioral Metrics (link)

Problem Statement: Customer churn is one of the costliest challenges in e-commerce, once a customer leaves, re-acquiring them costs 5x more than retaining them. This project applies logistic regression to predict which customers are likely to churn based on their behavioral and demographic data, enabling businesses to act before it's too late.

1.2. Logistic Regression

Why Logistic Regression?

Logistic regression was chosen because our target variable churn is binary. A customer either left or they didn't. Linear regression predicts a continuous number that can exceed 0 and 1, which is meaningless for a yes or no problem. We cannot tell a business that a customer has a 1.4 probability of churning. Logistic regression solves this by using the sigmoid function that S-shaped curve on the right which locks every prediction between 0 and 1. That output is a probability: specifically, how likely a customer is to churn.



Formula:

THE CORE FORMULA

$$P(\text{Churn} = 1) = 1 / (1 + e^{-z})$$

sigmoid function — squishes any value of z into a probability between 0 and 1

WHERE Z COMES FROM

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

linear combination of all your features, each weighted by a learned coefficient

WHAT EACH PART MEANS

β_0	$\beta_1 \ldots \beta_n$	$x_1 \ldots x_n$
Intercept — baseline churn probability when all features = 0	Coefficients — how much each feature increases or decreases churn odds	Feature values — Days_Since_Last_Purchase, Loyalty_Score, etc.

The model computes a weighted sum of all customer features, then wraps it through the sigmoid formula divided by 1 plus e to the negative z converting it into a clean, interpretable score. A customer scoring 0.82 gets flagged as high risk. A customer scoring 0.11 is predicted to stay. The decision boundary is 0.5, and that probability score is what turns this model into a ranked business priority list.

2. ETL / Data Feature Engineering

Variable Selection

The variables selected for the churn analysis focus on factors most directly related to customer retention and post-purchase behavior. Since the dataset includes customers who have already made purchases, the analysis should explain whether existing customers continue using the platform or stop engaging after purchase.

For this reason, the selected variables emphasize customer behavior, purchase patterns, loyalty, satisfaction, and engagement. Variables such as Purchase_Frequency, Days_Since_Last_Purchase, Loyalty_Score, Customer_Satisfaction, Return_Rate, Support_Tickets, Email_Open_Rate, and Social_Media_Engagement may help capture how actively customers interact with the platform and whether they are likely to churn.

Purchase-related variables such as Product, Quantity, Unit_Price, Discount_Applied, Payment_Method, and Payment_Delay_Days were also included because they may reflect buying habits and customer experience. Overall, the selected variables prioritize behavioral and retention-related signals rather than broad acquisition-focused factors.

New set of variables:

1	Customer_Segment
2	Annual_Income
3	Household_Size
4	Product
5	Quantity
6	Unit_Price
7	Discount_Applied
8	Payment_Method
9	Loyalty_Score
10	Loyalty_Tier
11	Customer_Satisfaction

12	Purchase_Frequency
13	Days_Since_Last_Purchase
14	Newsletter_Subscribed
15	Support_Tickets
16	Email_Open_Rate
17	Social_Media_Engagement
18	Purchase_Hour
19	Days_to_Next_Purchase
20	Return_Rate
21	Payment_Delay_Days
22	First_Purchase_Channel
23	Days_Since_Acquisition

+ **24. Churned**

ETL steps:

Extract	Transform	Load
Database Drops columns only keep 27 key features	Check duplicate Check missing value Encoding*	Final cleaned version of the data and load to the Logistic Regression Model

*Encoding:

- Logistic regression requires all inputs to be numeric, so categorical columns were converted using two approaches depending on whether their categories have a natural order

- Loyalty_Tier was **ordinal encoded**: mapped to 0,1,2,3 because the categories are ordered (Bronze, Silver, Gold, Platinum)
- The categorical columns (Customer_Segment, Product, Payment_Method, First_Purchase_Channel) have no order, so **one-hot encoding** was applied to create a binary column for each category
- drop=True was used to remove one redundant column per group and avoid multicolinearity

3. Model Results

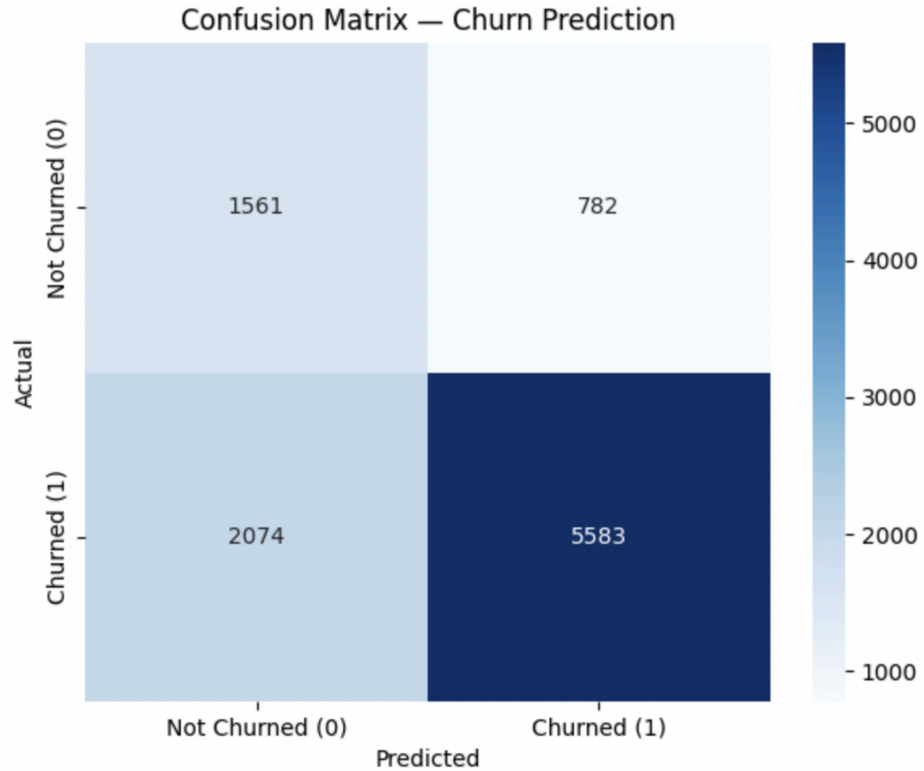
```
# Train & evaluate
model = LogisticRegression(class_weight='balanced', max_iter=1000,
random_state=42)
model.fit(X_train, y_train)

y_pred  = model.predict(X_test)
y_proba = model.predict_proba(X_test)[:, 1]

print(classification_report(y_test, y_pred))
print("AUC-ROC:", round(roc_auc_score(y_test, y_proba), 4))
```

```
             precision    recall  f1-score   support

           0       0.43      0.67      0.52      2343
           1       0.88      0.73      0.80      7657

    accuracy                           0.71     10000
   macro avg       0.65      0.70      0.66     10000
weighted avg       0.77      0.71      0.73     10000

AUC-ROC: 0.766
```

```
# Feature Importance

# Create the base dataframe
coef_df = pd.DataFrame({
    'Feature': X_train.columns,
    'Coefficient': model.coef_[0]
})

# Absolute values
coef_df['Abs_Coefficient'] = coef_df['Coefficient'].abs()

# Sort the values from most influential to least
coef_df = coef_df.sort_values('Abs_Coefficient', ascending=False)

print(coef_df)
```

```
                        Feature  Coefficient  Abs_Coefficient
5                 Loyalty_Score    -1.113144         1.113144
20          Customer_Segment_VIP    -0.098686         0.098686
24             Product_Product_E    -0.072973         0.072973
19      Customer_Segment_Regular    -0.062669         0.062669
26   Payment_Method_Digital Wallet     0.055236         0.055236
30   First_Purchase_Channel_Store     0.046305         0.046305
29   First_Purchase_Channel_Phone     0.045121         0.045121
28  First_Purchase_Channel_Online     0.044740         0.044740
7          Customer_Satisfaction     0.041586         0.041586
22             Product_Product_C    -0.030054         0.030054
11               Support_Tickets     0.018837         0.018837
1                 Household_Size     0.016265         0.016265
27         Payment_Method_PayPal     0.014538         0.014538
12               Email_Open_Rate    -0.012542         0.012542
25    Payment_Method_Credit Card     0.011812         0.011812
2                       Quantity    -0.011805         0.011805
0                  Annual_Income     0.010062         0.010062
18        Days_Since_Acquisition     0.009829         0.009829
23             Product_Product_D     0.009794         0.009794
16                   Return_Rate     0.009724         0.009724
8             Purchase_Frequency    -0.009345         0.009345
15         Days_to_Next_Purchase     0.008351         0.008351
3                     Unit_Price     0.008133         0.008133
6                   Loyalty_Tier     0.005335         0.005335
17            Payment_Delay_Days     0.004975         0.004975
4               Discount_Applied    -0.004834         0.004834
```

```
21          Product_Product_B     0.004778     0.004778
10        Newsletter_Subscribed  -0.003346     0.003346
14                Purchase_Hour  -0.002139     0.002139
9       Days_Since_Last_Purchase  -0.000758     0.000758
13       Social_Media_Engagement   0.000056     0.000056
```



4. Interpretation

The logistic regression model shows moderate predictive performance, with an overall accuracy of 0.71 and an AUC-ROC score of 0.766 (AUC-ROC is the preferred metric here given the class imbalance in the dataset). Since Churned is a binary variable where 1 indicates churn, the model performs better at identifying churned customers than non-churned customers. For churned customers, the model achieved a precision of 0.88, recall of 0.73, and F1-score of 0.80. This means that when the model predicts churn, it is often correct, although it still misses some churned customers.

Firstly, the confusion matrix evaluated the model on 10,000 unseen test customers. The stark color contrast of one dark navy box against lighter ones visually confirms what the metrics show: the model correctly identified 5,583 churners and 1,561 loyal customers, but missed 2,074 actual churners and over-flagged 782 non-churners. This imbalance stems from the **3:1 class ratio** in the dataset, meaning the model had far fewer loyal customer examples to learn from during training. To compensate, `class_weight='balanced'` was applied to prevent the model from defaulting to predicting churn for every customer.

In fact, the accuracy score shows precision of 0.88, recall of 0.73, and F1-score of 0.80. However, it missed 2,074 customers who actually churned, which represents the model's most significant limitation from a business perspective. Again, the model performs less strongly for non-churned customers, with a precision of 0.43 and F1-score of 0.52. **This suggests that the model has more difficulty correctly identifying customers who did not churn.**

Secondly, the feature importance results show that ***Loyalty_Score*** is the most influential variable. Its negative coefficient indicates that higher loyalty scores are associated with a lower likelihood of churn. Other influential variables include ***Customer_Segment_VIP, Product_Product_E, Customer_Segment_Regular, Payment_Method_Digital Wallet,*** and ***Customer_Satisfaction*****,** although their effects are much smaller.

Overall, the model suggests that churn is most strongly related to customer loyalty, with additional influence from customer segment, product type, payment method, and satisfaction. The model provides useful insight, but its overall performance is moderate rather than highly definitive.

5. Conclusion: Business Insights and Recommendations

This churn analysis used a logistic regression model because Churned is a binary outcome variable, where 1 indicates that a customer churned and 0 indicates that they did not. The model used customer behavior, purchase patterns, loyalty, satisfaction, and engagement-related variables to estimate churn likelihood.

Overall, the model showed moderate predictive performance, with an accuracy of 0.71 and an AUC-ROC score of 0.766. Because the dataset contained more churned customers than non-churned customers, the model may be better at identifying churned customers than non-churned customers, even though *class_weight='balanced'* was used to reduce this imbalance.

The most important predictor was Loyalty_Score, and its negative coefficient suggests that customers with higher loyalty scores are less likely to churn. Based on this, the company should focus on improving customer loyalty and retention by targeting low-loyalty or low-engagement customers with personalized offers, loyalty benefits, and proactive support.

Key actionable insights:

1. **Prioritize loyalty programs.** `Loyalty_Score` is the strongest predictor of retention in the model. Customers with high loyalty scores are significantly less likely to churn, which means investing in loyalty rewards, tiered membership benefits, and repeat-purchase incentives has a measurable protective effect on the customer base.
2. **Segment-based intervention.** The model identified `Customer_Segment_VIP` and `Customer_Segment_Regular` as influential variables. VIP customers who show early churn signals such as rising `Days_Since_Last_Purchase` should be prioritized for personal outreach, since losing a high-value customer carries disproportionate revenue impact compared to losing a new or occasional buyer.

3. Turn this business case report to an **Automated Weekly Customer Status Dashboard**.

   Use it to predict customer churn rate based on their X predictors.